Phase-II
Technology Stack (Architecture & Stack)

Date	02 March 2026
Team ID	Team 154
Project Name	Insurance Fraud Detection using ML
Maximum Marks	4 Marks

Technical Architecture Components:

This table outlines the specific technical building blocks of your application.

S.No	Component	Description	Technology
1.	**User Interface**	**Web-based form where investigators enter 24 claim parameters.**	**HTML5, CSS3, Jinja2 Templates.**
2.	**Application Logic**	**Backend server handling routing, data requests, and model triggering.**	**Python 3.x, Flask Framework.**
3.	**Data Preprocessing**	**Logic for handling missing values, scaling, and categorical encoding.**	**Pandas, NumPy, Scikit-Learn (StandardScaler).**
4.	**Machine Learning Model**	**Predictive engine trained on 12 different classification algorithms.**	**Scikit-Learn, XGBoost, CatBoost (saved as .joblib).**

5.	Model Serialization	Methodology for saving and loading the trained model for production.	Joblib library.
6.	Local Filesystem	Storage for the raw dataset and exported serialized model files.	Local Filesystem.
7.	Infrastructure	Environment where the application is hosted and executed.	Localhost (Development) / Render or AWS (Deployment).

Application Characteristics:

This section justifies the architectural choices based on industry-standard characteristics.

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Utilizes a complete open-source Python ecosystem for end-to-end ML.	Scikit-Learn, Flask, Pandas.
2.	Scalable Architecture	Uses a modular design separating the ML inference engine from the web frontend.	3-Tier Architecture (Client-Server-Model).
3.	Availability	Accessible via standard web browsers for remote insurance adjusters.	Web Browser (Chrome/Edge/Firefox).
4.	Performance	Optimized for real-time inference with a response time of <1 second.	Serialized Joblib Models.